

# Pratik Shirbhate

Lead AI/ML Platform Architect | MLOps & ML Platform Engineering | AWS, SageMaker, Terraform | Enabling Scalable Production ML Systems

■ Pune, Maharashtra, India

## CONTACT

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## SUMMARY

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I turn fragmented ML experiments into production-grade AI platforms — and I've spent 10+ years doing exactly that across healthcare, travel, and enterprise.

At Jet2, I lead ML Platform Engineering for a 50+ person data science organisation. In the last three years:

- Migrated the entire DS org from Dataiku to a cloud-native MLOps platform on AWS SageMaker — from architecture to go-live
- Cut cloud infrastructure costs by 50% (~£150K/year) through compute rightsizing, lifecycle policies, and VPC optimisation
- Productionised 20+ ML models across Retail Pricing, Marketing Attribution, Fuel Forecasting, and Revenue Risk
- Enabled self-service ML infrastructure for 50+ data scientists and ML engineers, reducing time-to-production significantly
- Built CI/CD, model governance, and observability frameworks from scratch — now the org standard

Before Jet2, I built predictive models in healthcare analytics at Evolent Health and developed my data engineering foundation at Cognizant.

**What I'm best at:** MLOps & ML Platform Architecture · AWS / SageMaker / Terraform · CI/CD for ML · Model Governance & Monitoring · Cost-Optimised AI Infrastructure · Cross-functional Engineering Leadership

I'm at my best when there's a messy, large-scale platform problem to solve — turning experimental chaos into reliable, industrialised AI.

Open to Principal / Staff / Lead roles in ML Platform Engineering, AI Infrastructure Architecture, or Enterprise AI Transformation. Based in Pune; open to remote or hybrid UK/India roles.

## PROFESSIONAL EXPERIENCE

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### Jet2 Travel Technologies Pvt Ltd. (5 years 7 months)

**Machine Learning Platform Lead** | April 2026 – Present | Pune

Led end-to-end migration of a 50+ member Data Science org from Dataiku to Amazon SageMaker, delivering a production-grade, multi-environment MLOps platform on AWS.

- Designed and architected cloud-native ML platform (SageMaker, Snowflake, Terraform, Azure DevOps) supporting 20+ ML projects across dev, test, and production
- Cut cloud infrastructure costs by 50%+ (~£150K/year) via compute rightsizing, lifecycle policies, and VPC optimization
- Established AWS account structure, IAM governance, and VPC architecture enabling secure self-service for 50+ users
- Standardized CI/CD pipelines, release governance, and Docker/ECR environments — increasing deployment frequency

- Built cost monitoring frameworks giving leadership first-ever visibility into per-project AWS spend
- Led incident management practices (PagerDuty, RCA) improving platform reliability
- Managed and mentored 5–8 ML engineers across UK and India, driving upskilling for 50+ engineers

#### **Technical Lead – Data Science & Machine Learning** | April 2023 – April 2026 | Pune

Led MLOps and CI/CD engineering, productionizing 20+ ML models across Retail Dynamic Pricing, Marketing Attribution, Fuel Forecasting, and Revenue Trading Risk.

- Founder of MLE Team; led 6–8 ML engineers, owning model productionization from experimentation handoff to live deployment
- Supported 40+ data scientists across business teams, establishing repeatable deployment patterns
- Built CI/CD pipelines for model training, validation, and deployment — standardizing release processes
- Established MLOps best practices: model monitoring, versioning, and lifecycle management
- Laid foundational ML infrastructure that directly evolved into the org's full-scale ML platform

#### **Senior Machine Learning Engineer** | April 2022 – April 2023 | Pune

- Built MLE Team from scratch; deployed scalable machine learning pipelines, transitioning models from research to production
- Developed automated workflows for model training, evaluation, and deployment using cloud-based infrastructure
- Collaborated closely with data scientists to productionize models and ensure performance, scalability, and reliability
- Implemented monitoring and logging mechanisms to track model performance in production
- Contributed to early MLOps practices, improving repeatability and maintainability of ML systems

#### **Senior Data Scientist** | November 2020 – April 2022 | Pune

- Developed and deployed machine learning models solving business problems across the Airline domain
- Performed data analysis, feature engineering, and model evaluation to improve predictive performance
- Collaborated with stakeholders to translate business requirements into data-driven solutions
- Delivered actionable insights and predictive models to support decision-making

### **Evolent Health (2 years 2 months)**

#### **Senior Data Scientist** | July 2020 – November 2020 | Pune

- Collaborated with US data science team and led India data science team
- Designed, developed, tested, and deployed data science solutions in healthcare analytics

#### **Data Scientist** | October 2018 – June 2020 | Pune

- Collaborated with US data science team to design, develop, and test components of data science solutions
- Built and deployed ML models for healthcare fraud detection, patient risk scoring, and claims processing

### **Cognizant (2 years 11 months)**

#### **Data Science and Big Data Associate** | November 2017 – September 2018 | Pune

#### **Data Science Associate** | November 2015 – November 2017 | Coimbatore, Tamil Nadu

- Collaborated with on-site lead data scientist to conduct data analysis and build proof of concepts for data science solutions

### **MAEER'S MIT College of Engineering**

#### **Assistant Professor** | July 2015 – November 2015 | Pune

- Taught Engineering Graphics, Fluid Mechanics, and Heat Transfer

## **TECHNICAL SKILLS**

MLOps & ML Platform Architecture · AWS SageMaker · Terraform · LLM Platform Engineering & GenAI Infrastructure · Infrastructure as Code · Snowflake · CI/CD for ML · Model Governance & Monitoring · Docker · Azure DevOps · Python · SQL · Apache Airflow · Apache Spark · GCP (GKE, Compute Engine) · Machine Learning (sklearn, H2O AutoML) · Deep Learning (TensorFlow) · Predictive Modeling · Feature Engineering · Tableau · MongoDB · Git · R

## KEY PROJECTS & BUSINESS IMPACT

### Jet2 – ML Platform & Infrastructure

| Project                                    | Impact                                                                                                                                                                                                                                                                                                                                                                                               |
|--------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Dataiku → AWS SageMaker Platform Migration | Led end-to-end migration of 20+ DS initiatives to a cloud-native MLOps platform (SageMaker, Snowflake, Terraform, Azure DevOps). Enabled self-service for 50+ users. <b>£150K/year cost savings</b> (50% reduction) via compute rightsizing, lifecycle policies & VPC optimization. Replaced legacy tooling with production-grade CI/CD, model governance, and observability — now the org standard. |
| MLE Team & MLOps Practice (Founded)        | Built the ML Engineering team from scratch (6–8 engineers). Productionized 20+ models across Pricing, Marketing, Fuel & Risk. Established repeatable deployment patterns for 40+ data scientists, standardized CI/CD pipelines, model monitoring, versioning & lifecycle management. <b>Reduced time-to-production significantly</b> and laid the foundation for the full-scale ML platform.         |

### Jet2 – Travel & Airline Revenue

| Project                                      | Impact                                                                                                         |
|----------------------------------------------|----------------------------------------------------------------------------------------------------------------|
| Hotel Price Elasticity – Surge Detection     | Real-time booking surge detection at hotel occupancy date level. <b>+5M GBP revenue</b> (2024, 20% properties) |
| Flight Price Elasticity – One Way Supplement | Real-time dynamic airline pricing model on Dataiku + GKE. <b>+3M GBP revenue</b> (2024)                        |
| Package Only Flights                         | Ensembled supervised, unsupervised & Pulp optimization models. <b>+30% Holiday Mix</b> for identified flights  |
| Customer Lifetime Value                      | KNN, K-Means, and RFM ensemble for targeted marketing (in development)                                         |

### Evolent Health – Healthcare Analytics

| Project                       | Impact                                                                                                                      |
|-------------------------------|-----------------------------------------------------------------------------------------------------------------------------|
| DRG Recoup (Fraud Detection)  | Ensembled XGBoost, GMM, IsolationForest, K-Prototype. <b>~3M USD annual savings</b> (1% fraud reduction)                    |
| Auto Intake (OCR Pipeline)    | OpenCV, Tesseract, NLP for Prior Authorization fax processing. <b>~4M USD annual savings</b> (40% manual effort reduction)  |
| Provider Matching             | Real-time ML API (AWS Lambda, TensorFlow) <300ms response. <b>~100K USD/year savings</b> (5% auto-adjudication improvement) |
| Readmission Prediction        | Automated ML pipeline (Airflow, sklearn, Keras, MongoDB). <b>3% fraudulent readmission reduction</b>                        |
| Medicaid Retention Model      | Logistic Regression, Random Forest, XGBoost + survival model for churn prediction                                           |
| Hospital Visit Classification | ML model for place-of-service prediction to optimize ED utilization. Deployed on Heroku via Flask                           |

| Project                       | Impact                                                                       |
|-------------------------------|------------------------------------------------------------------------------|
| Patient Risk Score Prediction | Risk scoring model for actuarial pricing & planning. Flask web app on Heroku |

## EDUCATION

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| Degree                                          | Institution                                | Year      | Score                           |
|-------------------------------------------------|--------------------------------------------|-----------|---------------------------------|
| M.Tech (Mechanical Engineering – Thermo-Fluids) | IIT Gandhinagar                            | 2013–2015 | 8.76/10                         |
| B.E. (Mechanical Engineering)                   | SGBAU, Maharashtra                         | 2009–2013 | 85% (4th Merit, College Topper) |
| HSC                                             | Board of Intermediate Education, Hyderabad | 2007–2009 | 90%                             |
| SSC                                             | Maharashtra State Board                    | 2007      | 84.76%                          |

## CERTIFICATIONS & ACCOMPLISHMENTS

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- AWS Certified Machine Learning Engineer – Associate
- Speaker at Analytics Vidhya Datahack Summit 2022
- Google Cloud Professional Machine Learning Engineer
- AI for Medicine Specialization (deeplearning.ai)
- AI for Medical Treatment
- Machine Learning (Udemy)
- Kubernetes (Simplilearn)
- Algorithms & Data Structures (Coursera)
- Programming, Data Structures and Algorithms Using Python
- Divide and Conquer, Sorting and Searching, and Randomized Algorithms
- Participation Certificate at Big Data Analytics Conference